Reg. No. | | | | | | | | | | |

[12EC55/13EC55]

FIFTH SEMESTER B.E/B.TECH EXAMINATION DECEMBER - 2015

MICROWAVE ENGINEERING

Time:3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions.
2. Missing data may be suitably assumed.

1. a. Define reflection co-efficient an derive an expression for reflection co-efficient at load in terms of load impedance. **06**

b. What are the properties of smith chart. **04**

c. Determine the load impedance and SWR. Given that the low loss line has a characteristic impedance of 500Ω and reflection co-efficient is $0.4\angle -50^{o}$. **04**

2. a. By considering elementary section of a transmission line derive equations for voltage and current at any point on a transmission line. **08**

b. A transmission line with characteristic 50Ω is terminated in a impedance of 25-j100Ω. Calculate the voltage reflection co-efficient and VSWR at the terminal load end of the line using smith chart. **06**

3. a. Using Helmholtz equation, derive the field equations for TE modes in rectangular wave guides. **08**

b. An Air-filled rectangular wave guide of inside dimensions a=7cm and b=3.5cm operates in the dominant TE_{10} mode. **06**

i. Find the cut-off frequency

ii. Determine velocity of the wave in the guide at a frequency of 3.5GHz.

iii. Determine the guided wavelength at the same frequencies.

4. a. Give the comparison between strip line and micro strip. **06**

b. Explain the characteristics of magic tee passive device, with a schematic diagram. Also obtain the s-matrix representation of the magic tee. **08**

5. a. Explain s-matrix representation of multiport network. **06**

b. With a neat sketch explain the four-port microwave circular and also obtain the s-matrix. **08**

6. a. Explain the construction, operation applications of reflex Kylstron. **10**

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b. A two cavity Klystron amplifier has the following characteristics **04**

Voltage gain = 15dB.

Input power = 5mWalt

Rsh of input cavity = 30 KΩ

Rsh of output cavity = 40 KΩ

Load impedance = 40 KΩ. Detemine

- **i.** Input rms voltage
- **ii.** Output rms voltage
- **iii.** Power delivered to the load.

7. a. With neat sketches, explain the IMPATT diode and draw the negative resistance curve. **08**

b. Explain the parametric amplifier with equivalence circuit. **06**

8. a. Explain the following attenuation measurement with neat diagram: **08**

- **i.** Power ratio method
- **ii.** RF substitution method.

b. Explain how impedance is measured using reflectometer with a basic block diagram. **06**